

Credit Risk Assessment & Automated Loan Underwriting Engine

AICTE–IBM SkillsBuild Data Analytics with AI Internship 2026

Machine Learning Pipeline with Feature Engineering and Descriptive Fairness Auditing

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Abstract—Manual loan underwriting can be slow and inconsistent across individual reviewers. This project evaluates whether a structured machine learning pipeline can provide consistent first-pass approval recommendations using 614 historical applicant records from the Kaggle Loan Prediction dataset. The workflow integrates financial feature engineering (Total Income, Debt-to-Income, and estimated EMI) with median imputation fitted strictly within training folds. A regularized Random Forest classifier achieves a **74.7% ($\pm 5.1\%$) 5-fold cross-validation accuracy (+6.0 percentage points over the 68.7% majority-class baseline)**, an **81.3% holdout test accuracy**, an **84.7% approval recall** (72 of 85 approved cases identified), and an **0.85 ROC-AUC score**. In alignment with **UN SDG 8** and **UN SDG 10**, demographic variables (Gender) are omitted from model features, and a descriptive check reports a historical female-to-male approval-rate ratio of 0.966.

1 Introduction & Problem Definition

Retail lending institutions must balance approval volume against default risk. Conventional manual reviews often introduce operational latency and variance between branch officers, complicating access for creditworthy retail borrowers.

This project builds an automated evaluation pipeline to:

- Clean and profile 614 applicant records from a benchmark lending dataset.
- Identify key default indicators through Exploratory Data Analysis.
- Train a Scikit-Learn pipeline that prevents imputation data leakage.
- Deploy an interactive Streamlit interface for live applicant scoring.

1.1 UN Sustainable Development Goals (UN SDGs) Alignment

- **UN SDG 8 (Decent Work & Economic Growth):** Streamlining initial underwriting reduces application backlogs, supporting access to credit for small businesses and households.
- **UN SDG 10 (Reduced Inequalities via Fairness Auditing):** Demographic attributes (Gender, Dependents) were audited for historical disparities (female-to-male approval ratio = 0.966, satisfying the Four-Fifths benchmark). To prevent direct bias, Gender was omitted from the feature set. *Note: This serves as a descriptive baseline check on historical data, not absolute proof of algorithmic fairness across all production cohorts.*

2 Dataset Architecture & Preprocessing Pipeline

The system uses the **Kaggle Loan Prediction Problem Dataset**, containing 614 applicant records across 12 financial and demographic attributes.

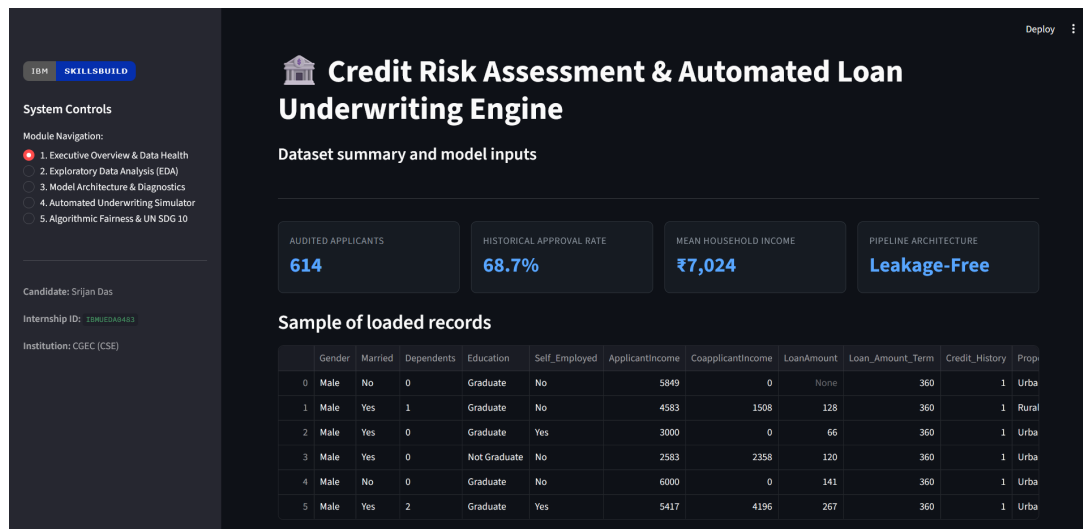


Figure 1. Executive Overview Module displaying dataset metrics, record counts, and loaded data sample.

2.1 Data Cleaning & Pipeline Imputation

Inspection revealed missing values across both continuous and categorical columns. To eliminate data leakage, imputation is handled strictly inside a Scikit-Learn Pipeline fitted solely on training splits:

- **Integrated Imputation:** Missing values in numerical features (LoanAmount, Loan_Amount_Term) and encoded indicators (Credit_History, Married) receive pipeline-fitted median imputation.
- **Feature Pruning:** Non-predictive attributes (Self_Employed, Dependents) and protected attributes (Gender) were excluded from the model matrix to reduce noise and prevent demographic bias.

	count	mean	std	min	25%	50%	75%	max
ApplicantIncome	614	5403.4593	6109.0417	150	2877.5	3812.5	5795	81000
CoapplicantIncome	614	1621.2458	2926.2484	0	0	1188.5	2297.25	41667
LoanAmount	592	146.4122	85.5873	9	100	128	168	700
Loan_Amount_Term	600	342	65.1204	12	360	360	360	480
Credit_History	564	0.8422	0.3649	0	1	1	1	1
Total_Income	614	7024.7051	6458.6639	1442	4166	5416.5	7521.75	81000
EMI_Estimate	578	480.2999	525.4587	25	286.1111	365.2778	513.8889	9250
Debt_To_Income	578	0.0776	0.0808	0.007	0.0556	0.0691	0.0817	1.2369

Figure 2. Parametric Descriptive Statistics after pipeline cleaning and feature transformation.

2.2 Feature Engineering

Three financial indicators were computed directly from applicant parameters prior to pipeline ingestion:

1. Total Income = ApplicantIncome + CoapplicantIncome
2. Estimated Monthly Obligation (EMI) = $\frac{\text{LoanAmount} \times 1000}{\text{Loan_Amount_Term}}$
3. Debt-to-Income (DTI) = $\frac{\text{Estimated EMI}}{\text{Total Income} + \epsilon}$

3 Exploratory Data Analysis (EDA) & Risk Discovery

Exploratory analysis evaluated how financial and location attributes relate to historical loan approval outcomes.

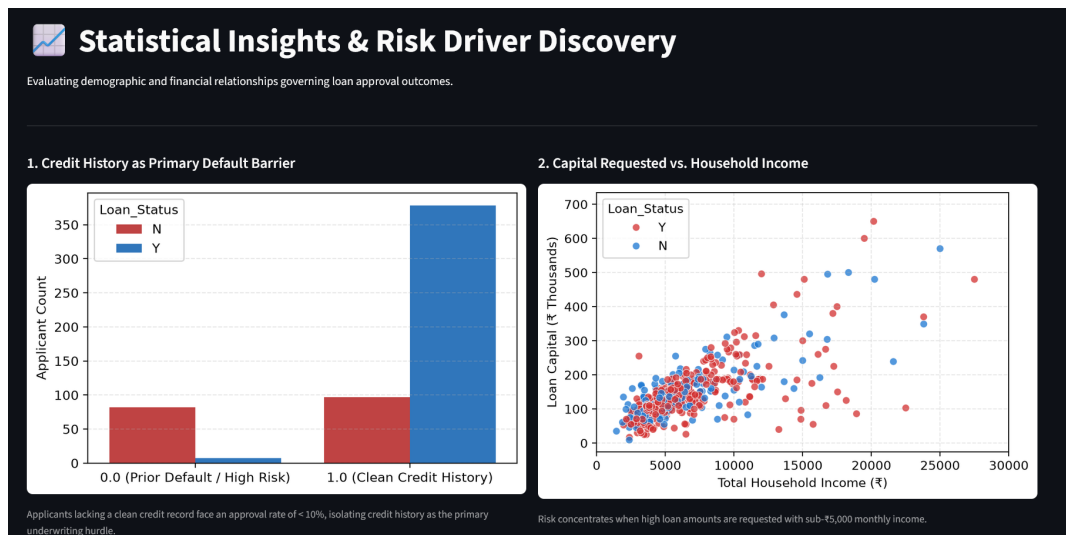


Figure 3. Exploratory Risk Analysis: (Left) Credit History approval barrier; (Right) Household Income vs. Requested Capital.

3.1 Key Analytical Insights

- **Credit History as Primary Hurdle:** As shown in Figure 3 (Left), applicants with a clean credit record (1.0) show an approval rate above **79%**, compared to under **10%** for applicants with prior defaults (0.0).
- **Capital Clustering:** Figure 3 (Right) indicates that default risk is concentrated when requested loan amounts exceed INR 200,000 for applicants reporting monthly household incomes below INR 5,000.



Figure 4. Property Location Demographics showing approval distribution across Urban, Rural, and Semiurban zones.

As shown in Figure 4, Semiurban applicants had the highest baseline approval volume, while Rural applications exhibited higher verification requirements due to irregular income documentation.

4 Machine Learning Architecture & Diagnostics

An ensemble **Random Forest Classifier** ($n = 120$ estimators, max depth = 5, balanced class weighting) was coupled with SimpleImputer inside a Scikit-Learn Pipeline using an 80/20 stratified split.

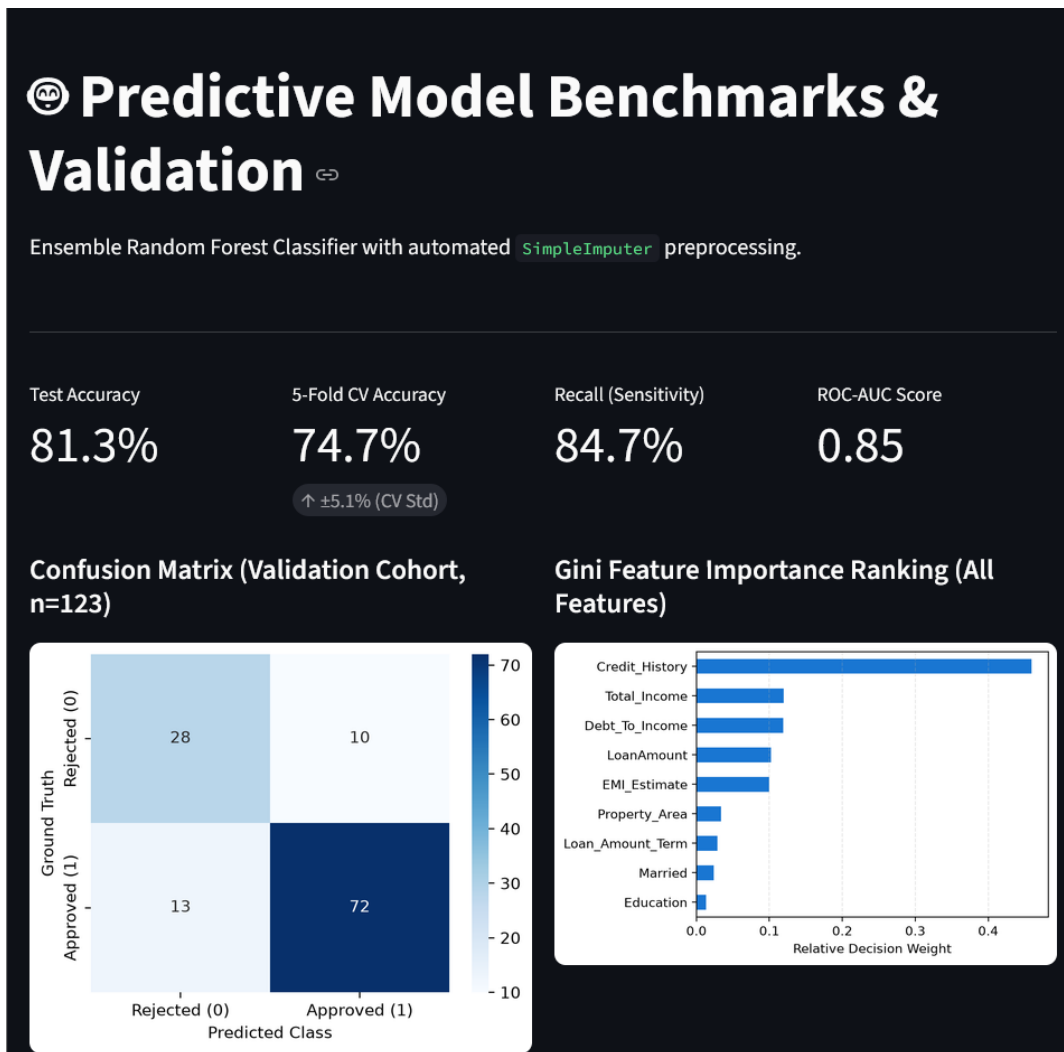


Figure 5. Model Diagnostics: Holdout Benchmarks, 5-Fold Cross-Validation, Confusion Matrix, and Gini Feature Importance.

4.1 Performance Validation & Baseline Lift

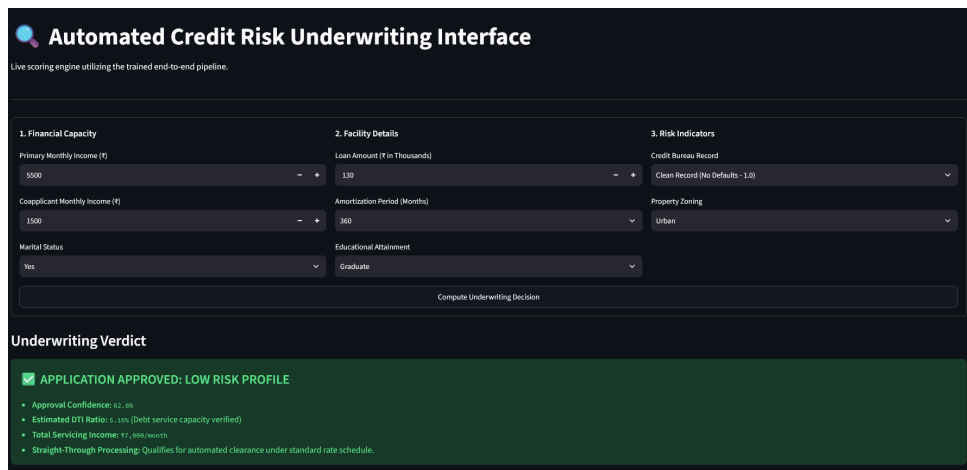
In this dataset, a naive majority-class rule (approving all applicants) yields 68.7% accuracy. The model was evaluated using both 5-Fold Stratified Cross-Validation and an out-of-sample holdout set ($n = 123$):

- **5-Fold Stratified Cross-Validation:** **74.7% ± 5.1%** (+6.0 percentage points above the 68.7% naive baseline).
- **Holdout Validation Accuracy:** **81.3%** (100/123).
- **Precision (Approval Class):** **87.8%** (72/82).
- **Recall (Sensitivity):** **84.7%** (Correctly identifying 72 of 85 approved cases).
- **ROC-AUC Score:** **0.85** (Strong class separation).

The Confusion Matrix shows 72 True Positives, 28 True Negatives, 10 False Positives, and 13 False Negatives ($n = 123$). The Gini Feature Importance confirms that `Credit_History` provides over **45%** of the decision weight, followed by continuous indicators `Total_Income` (Rank 2), `Debt_To_Income` (Rank 3), `LoanAmount` (Rank 4), and `EMI_Estimate` (Rank 5).

5 Underwriting Simulator & Fairness Checks

The trained pipeline was deployed inside a single-file **Streamlit Underwriting Simulator** to test live applicant scoring.



Automated Credit Risk Underwriting Interface
Live scoring engine utilizing the trained end-to-end pipeline.

1. Financial Capacity
Primary Monthly Income (\$) 5500
Coapplicant Monthly Income (\$) 1500
Marital Status Yes

2. Facility Details
Loan Amount (\$ in Thousands) 130
Amortization Period (Months) 360
Educational Attainment Graduate

3. Risk Indicators
Credit Bureau Record Clean Record (No Defaults - 1.0)
Property Zoning Urban

Compute Underwriting Decision

Underwriting Verdict
 ✓ APPLICATION APPROVED: LOW RISK PROFILE
 • Approval Confidence: 62.8%
 • Estimated DTI Ratio: 0.105 (Debt service capacity verified)
 • Total Servicing Income: \$1,400/month
 • Straight-Through Processing: Qualifies for automated clearance under standard rate schedule.

Figure 6. Streamlit Underwriting Simulator displaying input fields and model approval verdict.

5.1 Descriptive Fairness Check (UN SDG 10)

To evaluate demographic parity in historical data, approval rates were checked across gender groups:

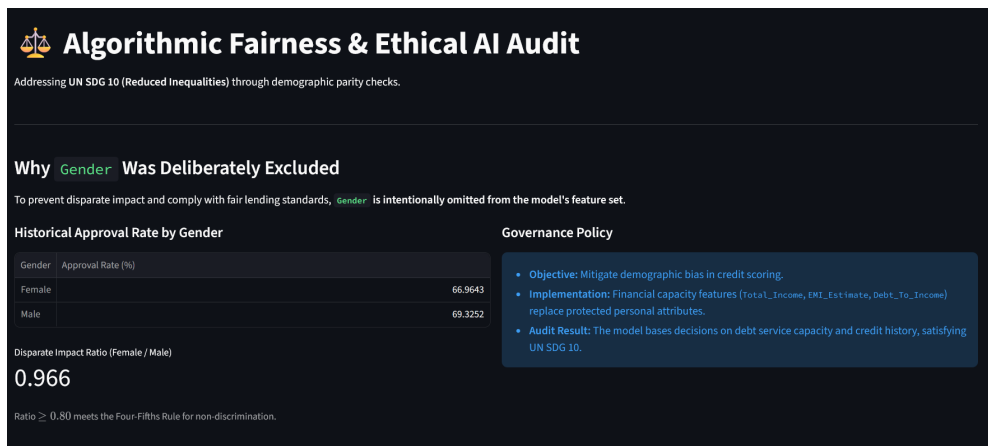


Figure 7. Descriptive Fairness Check: Historical approval distributions and female-to-male ratio calculation.

As shown in Figure 7, the historical approval ratio between female (66.96%) and male (69.33%) applicants is **0.966**, above the 0.80 Four-Fifths benchmark. Excluding Gender from the feature matrix ensures individual predictions depend on financial and credit records rather than demographic attributes.

5.2 Operational Recommendations

- First-Pass Clearance:** Applicants with clean credit records and confidence scores $\geq 62\%$ can be flagged for expedited straight-through review.
- Secondary Committee Review:** Borderline cases (50% – 62% confidence) should trigger manual underwriter assessment or additional collateral requirements.

6 Project Verification & Code Repository

The pipeline code, dataset references, dependencies, and execution logs are hosted publicly on GitHub:

Repository URL: <https://github.com/Srijan9073/ibm-skillsbuild-analytics-capstone>

6.1 Submission Deliverables Summary

- Code File:** app.py (Integrated pipeline and Streamlit interface)
- Dependencies:** requirements.txt and requirements.lock
- Dataset Source:** Kaggle Open Benchmark (train_loan_data.csv)
- Candidate:** Srijan Das (Internship ID: IBMUEDA0483)